

46-929, Fall 2019: Homework #3

Due 5:30 PM, Wednesday, November 13, 2019.

HW is to be turned in as a pdf file via Canvas only. Hand-written solutions are perfectly acceptable; of course, they should be clear.

Remember that you are to work **individually** on your homework assignments.

1. Do Problems 1, 2, and 3 of Section 13.8.1, on page 395, in Ruppert and Matteson. (That page is attached for your convenience.) For Problem 2, use AIC with `auto.arima()` to make your model selection.
2. For this exercise we will consider the data on three-month treasury bills described here:

<http://research.stlouisfed.org/fred2/series/TB3MS?cid=116>

Obtain the data, and restrict to the data from 1990 through 2012 (including both 1990 and 2012).

Please include all of your R code, along with all of the relevant plots. **Explain your steps clearly.**

- (a) Construct a plot of the original series. Does the series seem stationary?
- (b) Use differencing to find a transformation that results in an approximately stationary series. Remember that you do not want to do more differences than is necessary. **Justify your choice.**
- (c) Fit an ARIMA model to your series. Choose the values of p and q using AICc (corrected AIC). Set `approx = F` and `step=F` when you use `auto.arima()`.
- (d) Run diagnostic tests on the residuals from the above fit. Does the model seem to be a reasonable fit?
- (e) Forecast the next three months of the series. Create 95% prediction intervals for each.
- (f) You know the value of the series for those three months that you just forecast. Did the three intervals include the observed values?

13.7 Bibliographic Notes

Beran (1994) is a standard reference for long-memory processes, and Beran (1992) is a good introduction to this topic. Most of the time series textbooks listed in the “References” section discuss seasonal ARIMA models. For more details on HC and HAC covariance matrix estimators and the R package `sandwich` see Zeileis (2004). Enders (2004) has a section on bootstrapping time series and a chapter on multivariate time series. Reinsel (2003) is an in-depth treatment of multivariate time series; see also Hamilton (1994) for this topic. Transfer function models are another method for analyzing multivariate time series; see Box, Jenkins, and Reinsel (2008). Davison and Hinkley (1997) discuss both model-based and block resampling of time series and other types of dependent data. Lahiri (2003) provides an advanced and comprehensive account of block resampling. Bühlmann (2002) is a review article about bootstrapping time series.

13.8 R Lab

13.8.1 Seasonal ARIMA Models

This section uses seasonally non-adjusted quarterly data on income and consumption in the UK. Run the following code to load the data and plot the variable `consumption`.

```
1 library("Ecdat")
2 library("forecast")
3 data(IncomeUK)
4 consumption = IncomeUK[,2]
5 plot(consumption)
```

Problem 1 *Describe the behavior of consumption. What types of differencing, seasonal, nonseasonal, or both, would you recommend? Do you recommend fitting a seasonal ARIMA model to the data with or without a log transformation? Consider also using ACF plots to help answer these questions.*

Problem 2 *Regardless of your answers to Problem 1, find an ARIMA model that provides a good fit to `log(consumption)`. What order model did you select? (Give the orders of the nonseasonal and seasonal components.)*

Problem 3 *Check the ACF of the residuals from the model you selected in Problem 2. Do you see any residual autocorrelation?*

Problem 4 *Apply `auto.arima()` to `log(consumption)` using BIC. Which model is selected?*